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Quiet-STaR: Language Models Can Teach Themselves to Think Before Speaking

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Abstract

When writing and talking, people sometimes pause to think. Although reasoning-focused works have often framed reasoning as a method of answering questions or completing agentic tasks, reasoning is implicit in almost all written text. For example, this applies to the steps not stated between the lines of a proof or to the theory of mind underlying a conversation. In the Self-Taught Reasoner (STaR, Zelikman et al. 2022), useful thinking is learned by inferring rationales from few-shot examples in question-answering and learning from those that lead to a correct answer. This is a highly constrained setting – ideally, a language model could instead learn to infer unstated rationales in arbitrary text. We present **Quiet-STaR**, a generalization of STaR in which LMs learn to generate rationales at each token to explain future text, improving their predictions. We address key challenges, including 1) the computational cost of generating continuations, 2) the fact that the LM does not initially know how to generate or use internal thoughts, and 3) the need to predict beyond individual next tokens. To resolve these, we propose a tokenwise parallel sampling algorithm, using learnable tokens indicating a thought's start and end, and an extended teacher-forcing technique. Encouragingly, generated rationales disproportionately help model difficult-to-predict tokens and improve the LM's ability to directly answer difficult questions. In particular, after continued pretraining of an LM on a corpus of internet text with Quiet-STaR, we find zero-shot improvements on GSM8K (5.9%→10.9%) and CommonsenseQA (36.3%→47.2%) and observe a perplexity improvement of difficult tokens in natural text. Crucially, these improvements require no fine-tuning on these tasks. Quiet-STaR marks a step towards LMs that can learn to reason in a more general and scalable way.

"Life can only be understood backwards; but it must be lived forwards."

— Sren Kierkegaard

1 Introduction

Much of the meaning of text is hidden between the lines: without understanding why statements appear in a document, a reader has only a shallow understanding. Moreover, this has been repeatedly shown to be true for LMs as well, in the contexts of tasks ranging from commonsense reasoning to theorem proving to programming (Wei et al., 2022b; Nye et al., 2021; Zelikman et al., 2022; 2023a; Kojima et al., 2022). Reasoning about implications of text to predict later text has consistently been shown to improve LM performance on a variety of tasks, but methods for allowing LMs to learn from their reasoning (e.g., Zelikman et al. 2022) have focused on solving individual tasks or predefined sets of tasks (e.g., Wei et al. 2021b). These works rely on carefully curated datasets to provide either specific reasoning tasks or in some cases, the reasoning itself. We instead ask, if reasoning is implicit in all text, why shouldn't we leverage the task of language modeling to teach reasoning?

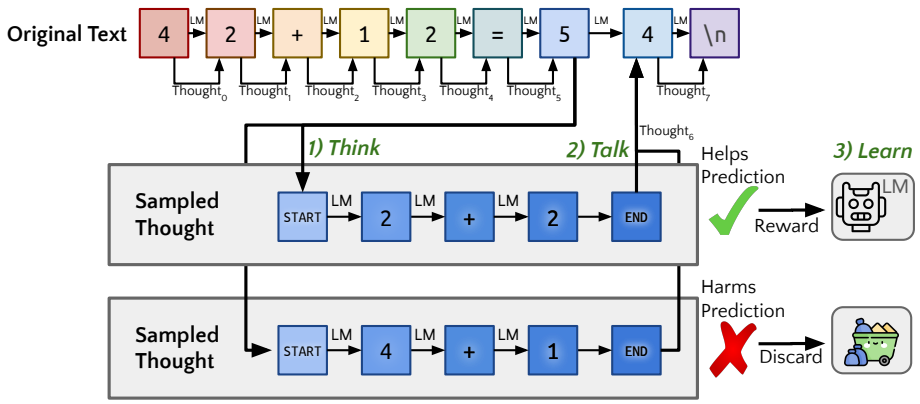


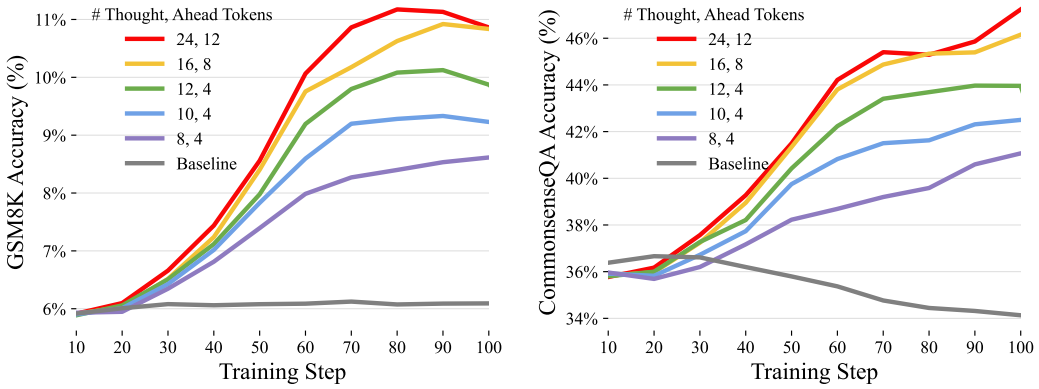
Figure 1: **Quiet-STaR**. We visualize the algorithm as applied during training to a single thought. We generate thoughts, in parallel, following all tokens in the text (think). The model produces a mixture of its next-token predictions with and without a thought (talk). We apply REINFORCE, as in STaR, to increase the likelihood of thoughts that help the model predict future text while discarding thoughts that make the future text less likely (learn).

In particular, the Self-Taught Reasoner (STaR, Zelikman et al. 2022) showed that LMs can bootstrap their reasoning ability on question-answering (QA) datasets by sampling rationales to attempt to answer questions, training on rationales if they led to a correct final answer, and then repeating this to iteratively solve more difficult problems. Yet, training from curated QA datasets limits the scale and generalizability of the rationales. QA datasets, especially high-quality ones, require thoughtful curation and will inherently only ever cover a subset of reasoning tasks. Thus, we extend STaR – instead of the LM learning to reason on particular tasks like mathematical QA, we train an LM to generate reasoning that helps it infer future text from a large internet text corpus. As a result, we allow the LM to learn from the diverse tasks present in language (Weber et al., 2021). This builds on an intuition essential to the current language modeling paradigm, namely, that "language models are unsupervised multitask learners" (Radford et al., 2019). Thus, as in STaR, we leverage the LM's pre-existing reasoning ability to generate rationales and train the LM on them with a REINFORCE-based reward (Williams, 1992). We refer to this technique as Quiet-STaR, as it can be understood as applying STaR "quietly", training the model to think before it speaks.

Broadly, Quiet-STaR proceeds by generating rationales after every token to explain future text (*think*), mixing the future-text predictions with and without rationales (*talk*), and then learning to generate better rationales using REINFORCE (*learn*). We apply Quiet-STaR to Mistral 7B (Jiang et al., 2023) using the web text datasets OpenWebMath (Paster et al., 2023) and Colossal Clean Crawled Corpus (C4, Raffel et al. 2020). We find that, even without dataset-specific fine-tuning, Quiet-STaR results in improvements to zero-shot direct-reasoning abilities on CommonsenseQA (36.3%→47.2%) and GSM8K (5.9%→10.9%), and that these improvements consistently increase with the number of tokens used in the LM's internal thoughts. Lastly, we qualitatively investigate patterns in the generated rationales.

In solving this task, we make the following contributions:

1. We generalize STaR to learn reasoning from diverse unstructured text data. To our knowledge, this is the first work explicitly training LMs to **reason generally** from text, rather than on curated reasoning tasks or collections of reasoning tasks.
2. We propose and implement a **parallel sampling algorithm** that makes our training procedure scalable, generating rationales from all token positions in a given string.
3. We introduce custom **meta-tokens** at the start and end of each thought to allow the LM to learn that it should be generating a rationale and when it should make a prediction based on that rationale.
4. We apply a **mixing head** to retrospectively determine how much to incorporate the next-token prediction from a given thought into the current next-token prediction.
5. We show that a **non-myopic loss**, including multiple tokens ahead for language modeling, improves the effect of thinking.
6. On multiple tasks, we demonstrate that thinking allows the LM to predict difficult tokens better than one trained on the same web text, improving with longer thoughts.



(a) GSM8K (b) CommonsenseQA

Figure 2: **Generalization Results**. We evaluate the extent to which the model trained with Quiet-STaR generalizes to directly answering problems that require reasoning. The left plot (a) shows the zero-shot accuracy on GSM8K, while the right plot (b) shows the zero-shot accuracy on CommonsenseQA, without any fine-tuning. In both plots, the x-axis represents training steps, and each line corresponds to a different number of thinking tokens used during Quiet-STaR training. The y-axis measures the zero-shot direct accuracy on the respective datasets. We also include an inference normalized version of this plot in Figure 6.

2 Related Work

2.1 Reasoning in Language Models

There have been many works on training and exploiting language models to solve difficult tasks by first training them to reason through them. For example, Rajani et al. (2019) demonstrated that a pre-trained language model fine-tuned to output on human reasoning traces before answering multiple-choice commonsense reasoning questions outperformed one trained directly on answers. Shwartz et al. (2020) demonstrated that language models, when provided with some scaffolding, can generate these helpful chain-of-thought solutions without additional supervision. Later, Nye et al. (2021) demonstrated that "scratchpads" required less scaffolding when the language models were more capable, a result later reinforced by Wei et al. (2022b), emphasizing informal tasks, and further strengthened by Kojima et al. (2022), demonstrating this behavior could be accomplished zero-shot. Most recently, Wang & Zhou (2024) showed further that for commonsense-question answering, one could force a language model to leverage chain-of-thought reasoning by preventing it from emitting any valid answer tokens unless it was confident. However, once again, these approaches only work for a question-answer dataset, and Wang & Zhou (2024) relies on heuristics to identify when the model has output answer tokens. Somewhat like TRICE (Phan et al., 2023), we use the relative improvements in the log-likelihood of the target text across rationales as an estimate of quality, but we simply subtract the mean reward and do not incorporate more complex control variates.

2.2 Training Language Models to Reason

One direction that researchers have used to train language models to reason or improve their reasoning is training the language model on mined reasoning traces or reasoning-like data (Rajani et al., 2019; Wei et al., 2021a; Lewkowycz et al., 2022; Chung et al., 2022; Gunasekar et al., 2023). Although this approach has been demonstrated to be effective, it comes with drawbacks. It requires either manual annotation, which is sensitive to the capability of the annotators and is off-policy for the language model (i.e., the distribution of reasoning is not text that the language model would otherwise likely have generated). This approach is also expensive, difficult to scale, and provides no clear path to solving problems harder than those that the annotators are capable of solving.

Another direction for teaching reasoning relies on a language model's own generated reasoning, which can be seen as building on a large body of literature on self-play (Silver et al., 2017; Anthony et al., 2017; Polu & Sutskever, 2020). These include methods such as the